REFERENCE IMPLEMENTATION / H_2 VQE

A two-qubit VQE reproduced the H_2 ground state.

A reproducible variational quantum eigensolver with independent exact-diagonalization and PySCF FCI validation. No hardware. No mock data.

ABSOLUTE ERROR VS EXACT DIAGONALIZATION

4.44×10^{-16} Ha

on the total H_2 energy — the same two-qubit Hamiltonian.

VQE	**−1.1373060357533986 Ha**
EXACT	**−1.137306035753399 Ha**

2	3	10 / 10
QUBITS AFTER TAPERING	UCCSD PARAMETERS	SCIENTIFIC CHECKS

H_2 · STO-3G · UCCSD · parity + Z_2 · exact statevector

The molecule is deliberately small. The validation is not.

Documented run · 2026-10-23 16:36 UTC

THE EXPERIMENT

Small system. Strict validation.

The toy molecule is deliberate: every intermediate result can be checked independently, from the fermionic Hamiltonian to the final JSON artifact.

MOLECULE 01

H_2 at 0.735 Å

STO-3G · 2 electrons · 4 spin orbitals

HAMILTONIAN 02

2 qubits · 5 Pauli terms

Parity mapping + two-qubit Z_2 tapering

ANSATZ 03

UCCSD · 3 parameters

Hartree–Fock reference · singles + doubles

OPTIMIZER 04

L-BFGS-B · 24 evaluations

4 iterations · converged · exact statevector

REFERENCES 05

Exact diagonalization + PySCF FCI

Same qubit Hamiltonian · independent implementation

EVIDENCE 06

10 / 10 checks passed

JSON artifact + 123 automated tests

THE 2-QUBIT HAMILTONIAN

$$H = -1.0523732458\, II + 0.3979374248\, IZ - 0.3979374248\, ZI - 0.0112801043\, ZZ + 0.1809311998\, XX$$

same Hamiltonian for every reference

SYSTEM DESIGN

One scientific layer. Two interfaces.

The Streamlit dashboard is a view over the validated quantum_vqe package — not a second implementation.

PIPELINE

PySCF integrals
geometry + STO-3G basis

↓

Fermionic Hamiltonian
second-quantized operators

↓

Parity + Z_2 tapering
4 spin orbitals → 2 qubits

↓

UCCSD + L-BFGS-B
3 parameters · 24 evaluations

↓

Independent validation
exact diagonalization + PySCF FCI

↓

Typed result artifact
energies · checks · timing · software

VALIDATED RESULT

−1.1373060357533986 Ha

10 / 10 checks · 123 tests

INTERFACES

CLI
`python -m quantum_vqe run-h2`

Streamlit
`streamlit run app.py`

Both call the same
`ExperimentConfig → quantum_vqe`

OPERATIONAL ENVELOPE

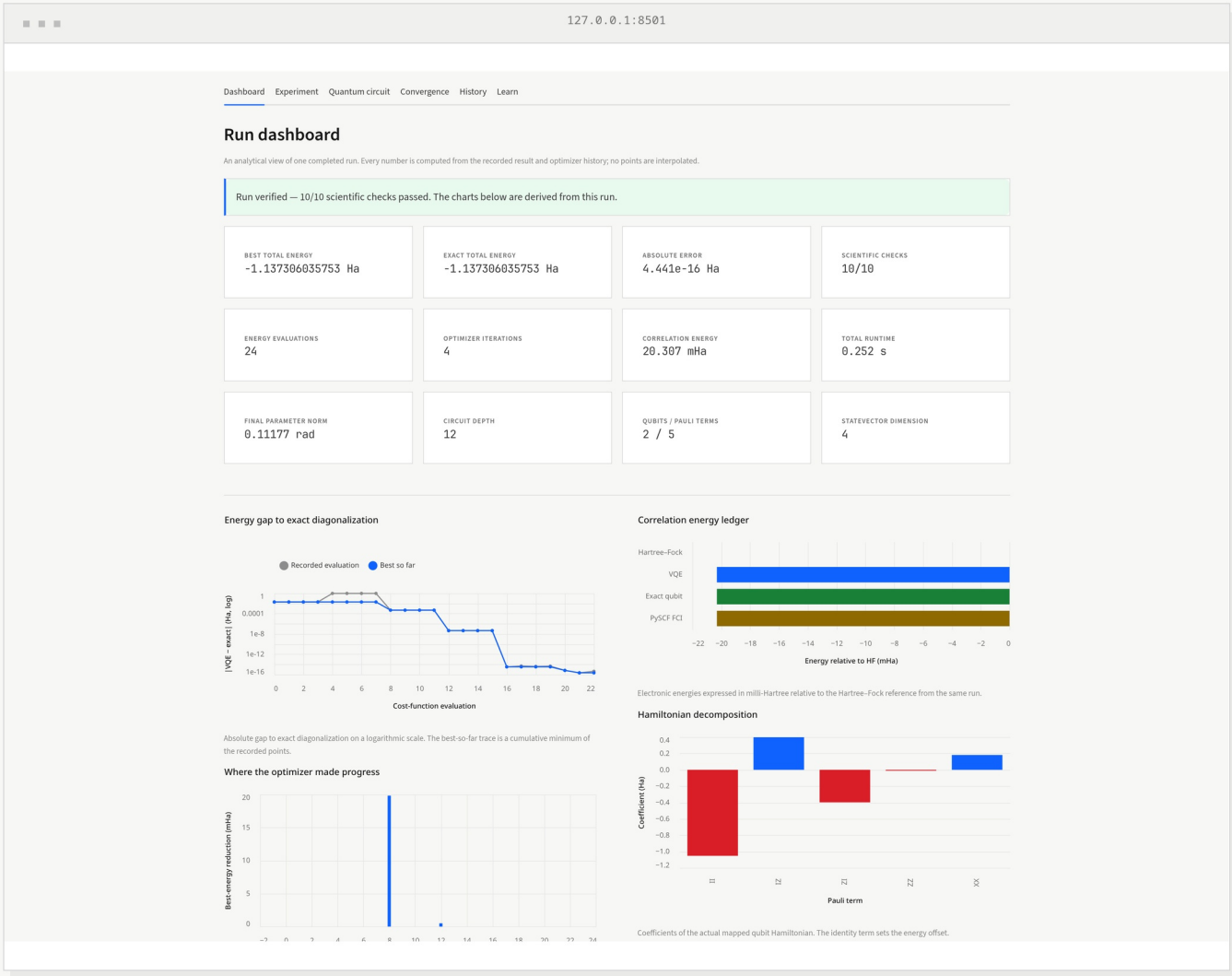
- Local CPU only
- Exact statevector · no shots
- No cloud account or API key
- Deterministic HF start
- Versioned JSON · MIT license

- **10 / 10 scientific checks passed**
  123 automated tests · CI 3.11 / 3.12

THE INTERFACE

A real workbench.

Configure, run, inspect the real-data dashboard, and export from a local Streamlit interface backed by the same scientific layer as the CLI.



STREAMLIT / LOCAL

No cloud · no API key · no hardware

DASHBOARD → RESULT → CIRCUIT

Every panel reads from ExperimentResult.

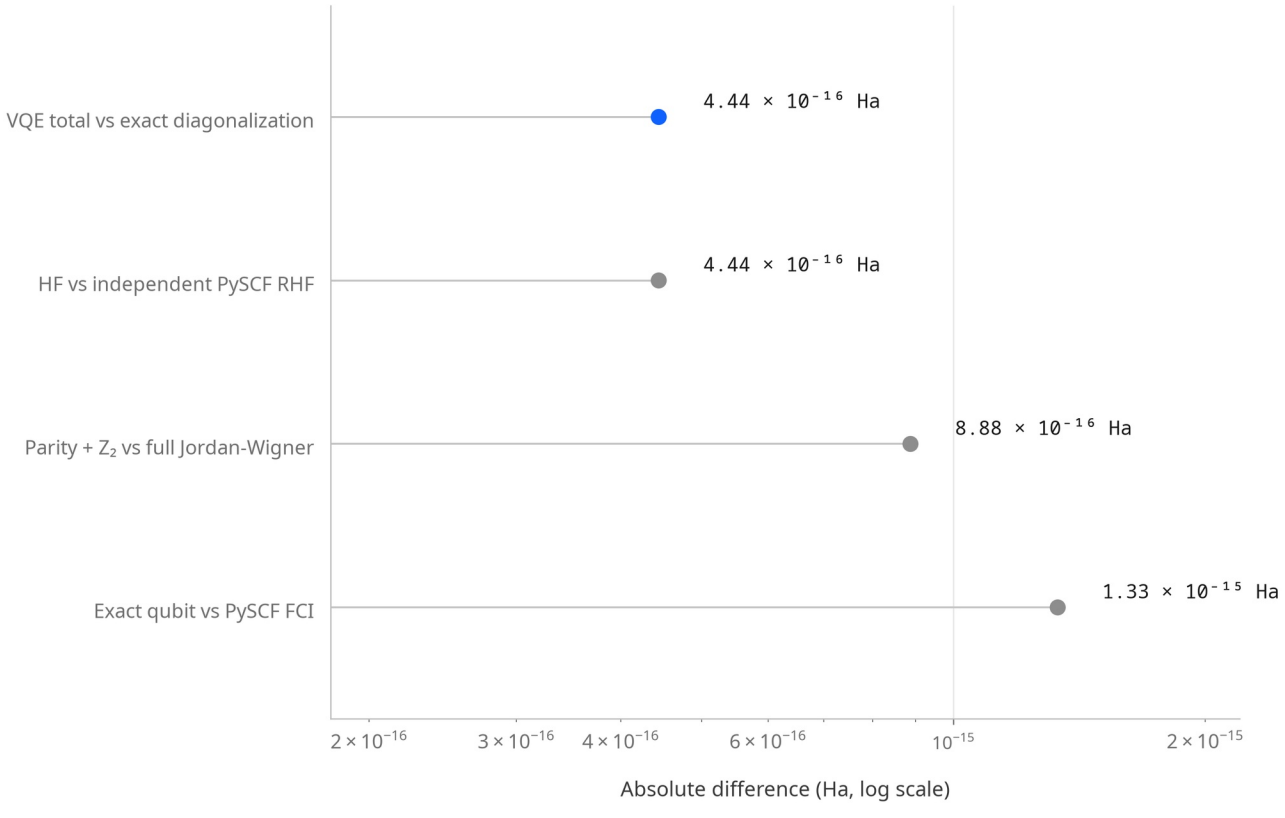
THE RESULT

Agreement to the last digit.

The VQE and exact diagonalization agree at 4.44×10^{-16} Ha on the total H_2 energy. Exact diagonalization and PySCF FCI agree at 1.33×10^{-15} Ha.

VQE TOTAL	EXACT QUBIT	ABSOLUTE ERROR
−1.1373060357533986 Ha	**−1.137306035753399 Ha**	**4.440892098500626 × 10⁻¹⁶ Ha**
optimized statevector	dense diagonalization	total energy

INDEPENDENT AGREEMENT CHECKS



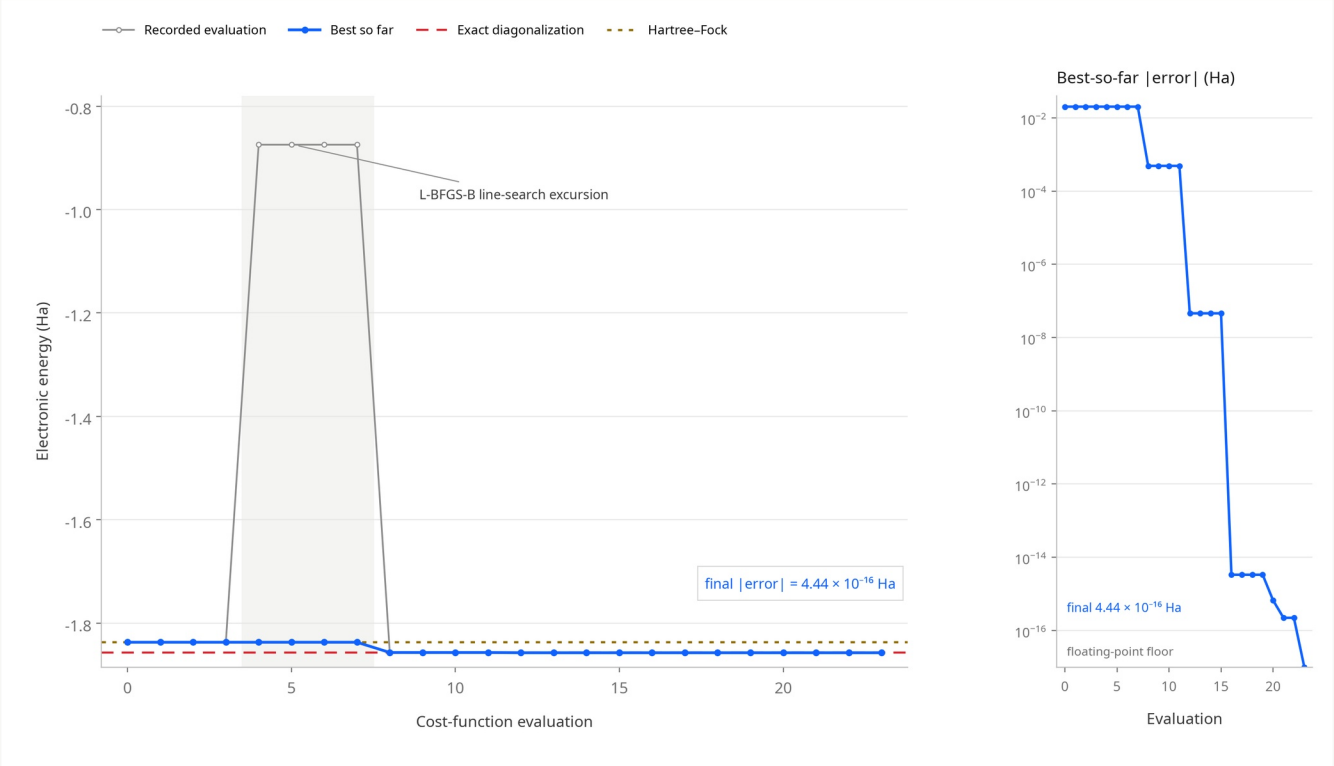
Every comparison lands at floating-point roundoff — no hand-entered energies.

10 / 10 checks passed

THE OPTIMIZER

What the optimizer actually did.

All 24 recorded cost-function evaluations, including the line-search excursion. No smoothing, interpolation, or synthetic points.



24
COST-FUNCTION EVALUATIONS

4
OPTIMIZER ITERATIONS

CONVERGED
L-BFGS-B STATUS

The excursion is real: L-BFGS-B evaluated higher-energy points during a line search. The best-so-far trace never moves backwards.

BEST-SO-FAR |ERROR|

4.44×10^{-16} Ha

FINAL VQE TOTAL

−1.1373060357533986 Ha

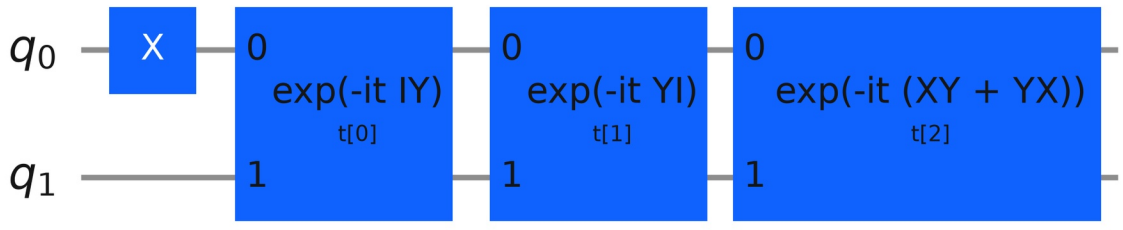
THE CIRCUIT

The ansatz is the executed object.

Rendered from the exact UCCSD circuit passed into the VQE — not a decorative illustration.

QUBITS	PARAMETERS	DEPTH · BASIS	GATES · BASIS
2	3	12	17

PAULI EVOLUTION / ONE DECOMPOSITION LEVEL



WHAT IS EXECUTED

The Hartree–Fock reference is prepared, then UCCSD excitation blocks act on two qubits. The estimator evaluates the same five-term Hamiltonian on the resulting statevector.

WHY IT MATTERS

Depth and gate counts are computed from the same circuit object used by the run. The dashboard and CLI cannot drift into different circuit claims.

BASIS-GATE COUNTS

H 4 · CX 4
RY 2 · SX 2
RZ 2 · U 1
SXDG 2

Basis-gate decomposition available in the dashboard.

UCCSD · 3 parameters · 2 qubits · 17 basis gates

REPRODUCIBILITY

The result is the easy part. The evidence is the product.

The repository ships the experiment, the dashboard, the tests, and the exact artifact that produced the headline number.

123

AUTOMATED TESTS

unit + integration, including a real H_2 run

10 / 10

SCIENTIFIC CHECKS

hermiticity, convergence, references, taper cross-check

3.11 / 3.12

CI MATRIX

green on GitHub Actions

MIT

LICENSE

open source · local · no cloud or hardware

RUN IT YOURSELF

Read the code. Reproduce the result. Challenge the claims.

github.com/Abdelhakim-Baalla/quantum-vqe-chemistry-lab

MIT licensed · Python 3.11 / 3.12 · no quantum hardware required

H_2 / STO-3G / UCCSD / 2Q